



AuRa dataset: A vision dataset from a bike's perspective for autonomous robots in urban environments

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Received: 29 January 2025 / Accepted: 6 October 2025
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Abstract

In the field of autonomous robotics on public roads, accurate perception of the environment is crucial for safe and efficient operation. In particular, the training of algorithms for object detection and semantic segmentation requires datasets that can map the complex visual scenes in road traffic. Open access datasets for autonomous driving are primarily recorded with street vehicles and therefore only contain scenes in structured traffic areas but not in areas for cyclists and pedestrians. In this paper, we present a dataset for object detection and semantic segmentation that is specifically tailored to the particular challenges of autonomous robots in urban environments. Our dataset, recorded from the perspective of a bicycle, is based on real-world scenarios and different environments and includes a wide range of objects commonly encountered in urban environments. To ensure robustness and generality, our dataset includes hand-labeled annotations at a high level of detail, providing precise bounding boxes around each object of interest, and semantic segmentation masks at pixel level and comprehensive ground truth labels for training and evaluation. Our dataset will serve as a valuable resource for researchers, engineers and developers working on autonomous robotics, enabling advancements in object detection algorithms and fostering the development of safer and more efficient autonomous robot systems in urban scenarios.

Keywords Deep learning vision dataset · Urban robotics · Autonomous driving

1 Introduction

Autonomous robots and micro-mobility vehicles have the potential to positively impact the environment and contribute to climate change mitigation. These vehicles can effectively replace cars when equipped with automated functions, helping to reduce emissions and traffic congestion (Schmidt et al. 2021). However, while existing datasets support the development of perception systems for autonomous vehicles, there is a lack of data that reflects the unique conditions faced by robots and autonomous systems operating in bicycle-dominant or mixed-use environments. To address this gap, we created a novel dataset tailored for autonomous systems that operate in bicycle-focused infrastructure. Our dataset captures the world from the viewpoint of an autonomous bicycle, focusing on scenarios and objects that are typically underrepresented in other datasets. This includes bicycle lanes, pedestrian pathways, parked bicycles, and other objects relevant to bicycle paths and mixed-use zones. Our contributions include detailed annotations for both object detection and semantic segmentation, capturing critical elements such as cyclists with or without their bicycles,

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pedestrian interactions, and infrastructure elements like bicycle lane boundaries, vertical barriers, and curb stones. These annotations are essential for developing perception models that allow mobile robots to make safe and efficient decisions in complex, unstructured environments. By providing a training basis rooted in the perspective of an autonomous cargo bike, we aim to drive advancements in urban mobility and robot perception, paving the way for more capable and versatile autonomous systems.

2 Related work

Several public datasets have made important contributions to the development of autonomous driving and scene understanding. KITTI (Geiger et al. 2012), Cityscape (Cordts et al. 2016), Nuscenes (Caesar et al. 2020), Waymo Open Dataset (Sun et al. 2020), and the View-of-Delft Dataset (Palffy et al. 2022) are among the most widely used, but they are primarily designed for vehicle-based perception and general object recognition in structured road environments. These datasets, while valuable, often fail to capture the specific perspectives and challenges encountered by autonomous robots or vehicles that navigate on bicycle paths or in pedestrian-dominated spaces. To broaden the scope of perception beyond car-centric views, several datasets have been introduced. The Mapillary Vistas dataset (Neuhold et al. 2017) offers a large-scale, finely annotated set of street-level scenes, including various urban objects such as pedestrians and bicycles. The X-World dataset (Zhang et al. 2021) aims to enhance scene understanding across multiple viewpoints, including non-vehicle perspectives. SideGuide (Park et al. 2020) targets pedestrian and bicycle path segmentation and directly addresses perception challenges in micro-mobility contexts. SANPO (Waghmare et al. 2024) focuses on robot navigation in pedestrian-oriented environments, aligning closely with the needs of delivery robots and autonomous bicycles. Space-awareVLM (Han et al. 2025) introduces a

spatially-aware visual-language dataset for understanding urban spatial relations. DensePASS (Ma et al. 2021) provides densely labeled panoptic segmentation for detailed scene understanding in non-car-centric settings. TAS500 (Metzger et al. 2021) focuses on perception in unstructured driving scenarios, while the FDWW dataset (Li et al. 2016. BDD100K (Yu et al. 2020) is another large-scale, diverse driving dataset that covers a wide range of weather conditions, times of day, and driving scenarios across multiple U.S. cities. It supports a variety of tasks, including object detection, semantic segmentation, and lane marking. While BDD100K greatly improves the diversity and scale of driving data, its perspective remains predominantly car-centric and focused on roadway navigation. Additionally, the OPT-micro dataset (Sergey 2025) is currently in development and aims to provide a multimodal, micromobility-focused dataset recorded from the perspective of an autonomous microvehicle. Although these datasets represent significant progress, none specifically address the challenges faced by autonomous platforms operating from the perspective of a bicycle or micromobility device. Our dataset seeks to fill this gap by providing high-quality, fine-grained annotations from this unique and underrepresented viewpoint.

3 Data Acquisition

Our recording platform (Fig. 1) is a three-wheeled cargo bike equipped with sensors and actuators to enable autonomous driving (Schmidt et al. 2021) and real-time trajectory prediction (Sass et al. 2023). This setup includes three cameras, two laser scanners, an IMU, GPS, a steering angle sensor, and a wheel speed sensor, collecting over 45 hours of raw sensor data in the city of Magdeburg, Germany—a Central European city with around 240,000 inhabitants. For this dataset, only camera data is utilized, featuring one front camera for long-range views and two side cameras focused on lateral areas, providing robust coverage of urban

Fig. 1 Sensor platform used for recording the raw sensor data for the dataset. Left: side view; Right: front view with cameras and sensors



Table 1 Technical specifications of the cameras

Orientation	Front	Side
Camera model	Basler acA 2000 - 50gc	Basler a2A 1920 - 51gcBAS
Lens model	Basler C10-1214-2M-S	Edmund Optics 3.5MM COMPACT
Resolution [px]	2046x1086	1936x1216
Colour depth [bit]	12	12
Mounting height [mm]	750	710/790 (right/left)

environments. Table 1 outlines the camera specifications. To enhance model resilience in real-world applications, we included images of varied quality, from high-quality captures to those affected by motion blur or low lighting. This approach ensures that models trained on our dataset are capable of handling challenging conditions in real-world scenarios.

Data collection spanned various seasons (Fig. 2) and routes in diverse areas of Magdeburg, covering typical urban environments such as the city center, inner-city residential areas, shopping districts, and a university campus. The dataset includes four primary driving domains: parks and green areas, pedestrian zones and traffic-calmed areas, bicycle/pedestrian paths, and car roads (Fig. 3). Each domain captures unique urban scenarios and object distributions:

- Parks and Green Areas (Fig. 3a) account for a smaller share of bounding boxes (4.45% front, 4.25% side) but have higher semantic segmentation at 17.74%, emphasizing navigation in open, natural spaces, including locations like parks and playgrounds.
- Pedestrian Zones and Traffic-Calmed Areas (Fig. 3b) include sidewalks and pedestrian-only spaces. These areas contain 11.53% front and 18.15% side bounding boxes,

with 21.05% semantic segmentation, reflecting areas with high pedestrian density and various obstacles. The side cameras often capture views of these zones, including marketplaces and other pedestrian-centric areas.

- Bicycle/Pedestrian Paths (Fig. 3c) contribute the largest category, with 64.56% front, 60.18% side bounding boxes and 39.77% semantic segmentation. This category encompasses a range of paths, including bike lanes adjacent to streets, paths that also serve pedestrians, and dedicated bike roads, highlighting the detection needs in mixed-use areas.
- Car Roads (Fig. 3d) are prominent, with bounding boxes at 19.46% front and 17.41% side, and semantic segmentation at 21.44%. This category covers standard road settings, supporting the development of autonomous capabilities in vehicle-focused environments.

Additionally, the dataset captures several specific environment types, such as bicycle paths beside streets (also covering sidewalk areas), traffic-calmed streets, playgrounds, and markets. These detailed scene categorizations provide a comprehensive basis for developing robust perception algorithms across varied urban scenarios.

4 Data description

The dataset comprises color images annotated for both bounding boxes and semantic segmentation. Figure 4 illustrates the dataset's directory structure. Semantic segmentation annotations are stored as 8-bit grayscale PNG images, while bounding box annotations are provided in XML format. Additionally, auxiliary scripts for visualization are included to facilitate interpretation. Annotations

Fig. 2 Left: percentage distribution of the recording periods, right: percentage distribution of driving domains

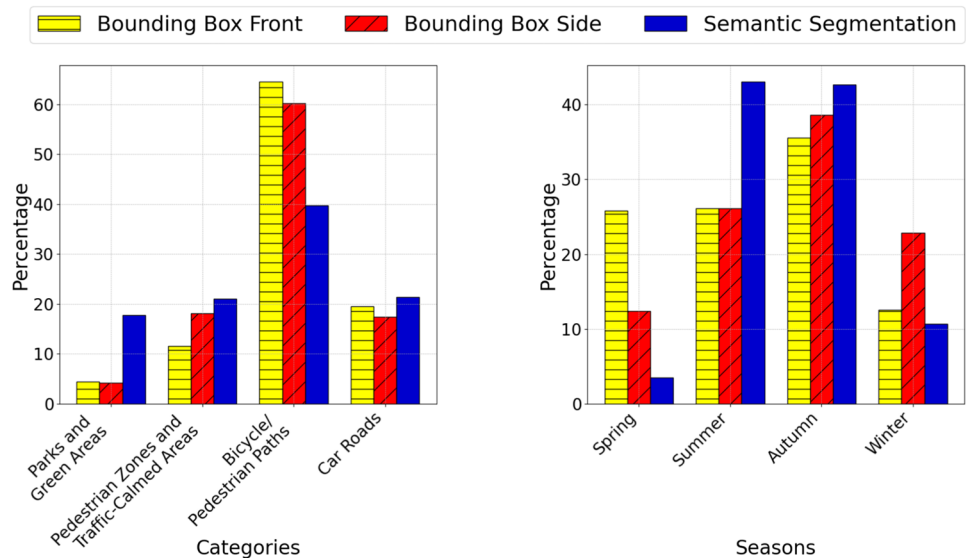


Fig. 3 Examples of driving domain categories



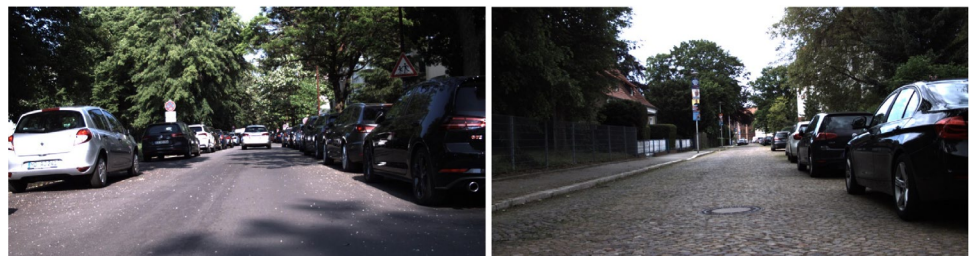
(a) *Parks and Green Areas*



(b) *Pedestrian Zones and Traffic-Calmed Areas*



(c) *Bicycle/Pedestrian Paths*



(d) *Car Roads*

were performed using the annotation tool *Acotate* www.acotate.ai, developed by FusionSystems GmbH. The dataset also includes a training directory, which contains scripts for data loading, training, and evaluating baseline models for bounding box detection and semantic segmentation. The trained models are saved in their respective subdirectories. Furthermore, the *utils* directory provides helper scripts for visualization, such as rendering bounding boxes and displaying semantic segmentation masks, enabling efficient data inspection and result analysis.

4.1 Object detection annotations

Objects were annotated using bounding boxes on two distinct datasets, captured from different camera perspectives:

two side cameras and one front camera. The division into front and side views further enhances the dataset's robustness and ensures that models can recognize objects from multiple perspectives. The side camera dataset consists of 3,611 images with a total of 19,096 annotations, while the front camera dataset contains 2,738 images with 23,188 annotations. The annotations include a variety of object classes relevant to our detection task. Each image was manually annotated to ensure high quality annotation with an accuracy of three pixels. Figure 5 shows examples of object detection using bounding boxes. The boxes highlight different classes with annotations corresponding to each detected object. Detailed information on the number of annotations per object class for both datasets is provided in Table 2, which outlines the frequency distribution of each annotated

```

aura_dataset
|-- dataset
|   |-- bb
|   |   |-- {train_validate,test}/
|   |   |   |-- {front,side}/
|   |   |   |-- annotations
|   |   |   |-- images
|   |-- semantic
|   |   |-- {train_validate,test}/
|   |   |   |-- {front,side}/
|   |   |   |-- annotations
|   |   |   |-- images
|-- training
|   |-- bb
|   |-- semantic
|-- utils
|-- README.md

```

```

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    <height>1086</height>
    <depth>3</depth>
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    <name>car</name>
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    <truncated>0</truncated>
    <difficult>0</difficult>
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      <ymin>442</ymin>
      <xmax>2045</xmax>
      <ymax>758</ymax>
    </bndbox>
  </object>
</annotation>

```

Fig. 4 Left: structure of the provided dataset, right: bounding box annotation format

Fig. 5 Examples of object detection. Each image shows the original scene with annotated bounding boxes for various objects



object. The objects are categorized into various classes based on common urban elements such as vehicles, cyclists and traffic-related infrastructure.

- *Vehicles* are a central class for object detection in urban environments. The dataset provides extensive annotations for different types of vehicles. Cars, the most common vehicle, have 8,635 front-facing and 7,775 side-facing annotations. Transporters and trucks, which are larger vehicles, have comparatively fewer annotations, with 1,035 and 935 transporters, and 295 and 219 trucks annotated from front and side views, respectively. Other vehicles, such as buses and trams, are less frequent but still covered, with trams having 417 front and 318 side annotations. Emergency vehicles, including

ambulances, police vehicles, and fire trucks, have limited but distinct annotations. These annotations capture the diversity of vehicles encountered in traffic.

- *Cyclists and Motorcycles*, both with and without their respective riders, form a critical part of the dataset. Cyclists with bicycles are annotated with 1,343 front and 562 side views, while bicycles without cyclists have 851 and 1,326 annotations for front and side perspectives, respectively. This distinction helps in detecting whether a bicycle is actively being used or parked. Motorcycles follow a similar pattern, with separate annotations for motorcycles with and without motorcyclists. Although the number of motorcyclists is small, this division is essential for accurately identifying these objects in dynamic environments.

Table 2 Number of annotated objects for object detection

Class	Subclass	Front	Side	Sum
Vehicles	Car	8635	7775	16,410
	Transporter	1035	935	1988
	Tram	417	318	735
	Truck	295	219	514
	Bus	50	25	75
	Police vehicle	24	22	46
	Ambulance vehicle	7	12	19
	Fire truck	1	0	1
				19,788
Cyclists	Cyclist with bicycle	1343	562	1905
	Bicycle without cyclist	851	1326	2177
	Motorcycle without motorcyclist	16	41	57
	Motorcyclist with motorcycle	4	8	12
				4151
Persons	Adult without baby buggy or mobility support	5063	4563	9626
	Person with baby buggy	194	191	385
	Person with mobility support	128	127	255
	Child without baby buggy or mobility support	40	79	119
				10,385
Animals	Dog	31	25	56
Traffic-Related Infrastructure	Traffic sign	3948	2261	6209
	Traffic light	1001	331	1332
	Traffic markings	103	254	357
	Speed bump	2	4	6
	Signalling device railroad crossing	0	1	1
				7905
		23,188	19,096	42,284

The values highlighted in bold are the sum of all subclasses or the corresponding main class

- The *Persons* category includes diverse subclasses, such as adults, children, and individuals using mobility aids or pushing baby buggies. These distinctions are important for object detection models that must understand the varying sizes and characteristics of different types of pedestrians. With nearly 10,000 annotations of adults and several hundred instances of other pedestrian types, this class is crucial for developing pedestrian-aware systems that prioritize human safety in complex environments.
- *Animals*, including dogs, are less frequent in the dataset but have 56 annotations in total, reflecting their common presence in both urban and suburban areas. This class ensures that object detection models can respond appropriately to small, moving obstacles like pets.
- *Traffic-Related Infrastructure* includes elements like traffic signs, traffic lights, and traffic markings, which are essential for navigation and obeying road rules. With over 6,000 instances of traffic signs and over 1,300 instances of traffic lights, these annotations provide ample data for recognizing critical infrastructure. Less frequent but still important elements like speed bumps and railroad crossing devices are also annotated, ensuring the dataset captures a wide variety of road environments.

4.2 Comparison with existing object detections datasets

Table 3 summarizes the comparison between our dataset and several established object detection datasets, including COCO, KITTI, Cityscapes, and BDD100K.

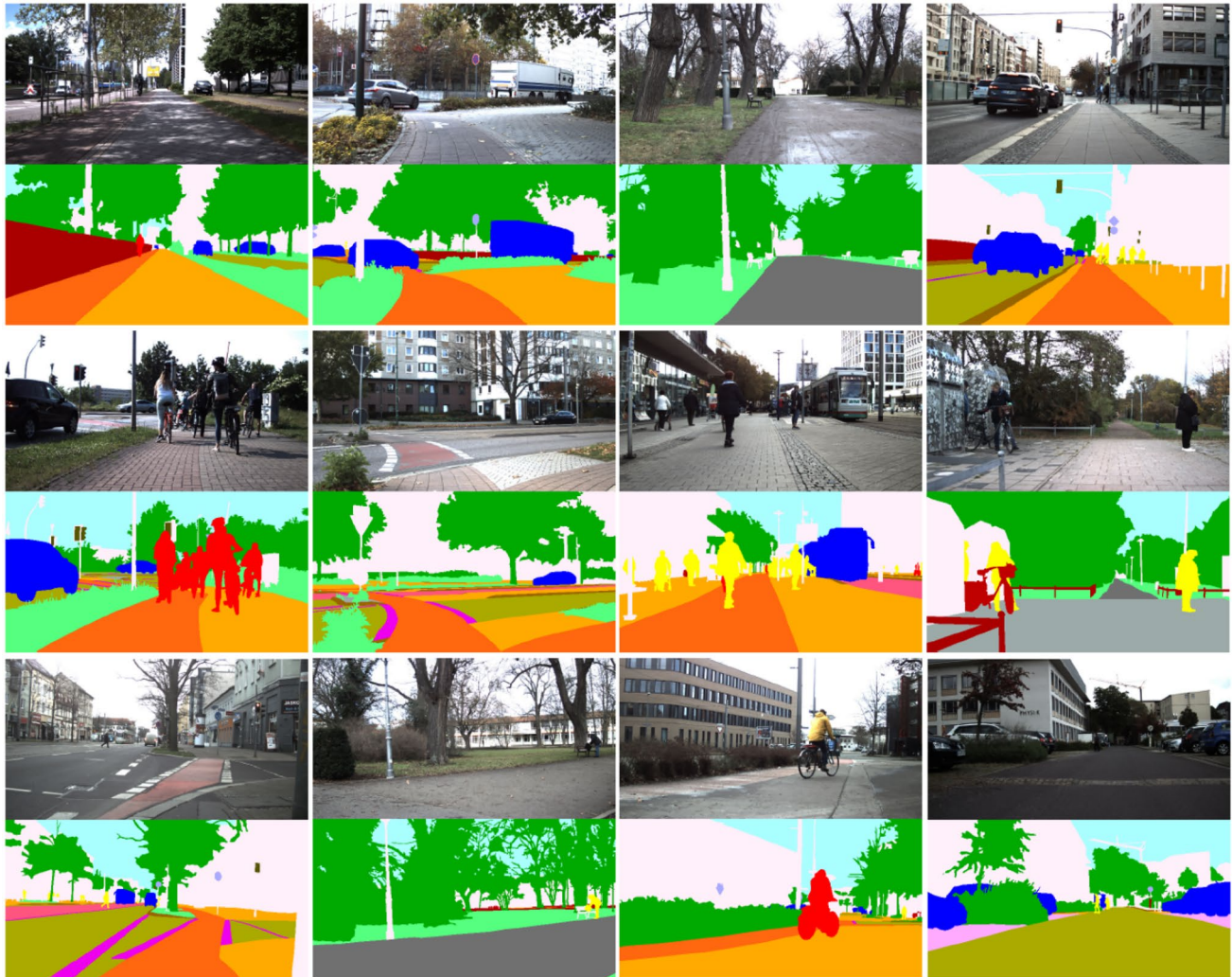
The comparison highlights that our dataset offers a larger and more diverse annotation set for its size, including different perspectives (front and side views), detailed pedestrian and cyclist subclasses, and rare object categories such as emergency vehicles and traffic infrastructure. These characteristics make it particularly valuable for urban robotics and micromobility applications.

4.3 Semantic segmentation annotations

For semantic segmentation, we utilized only the front camera images, resulting in a dataset of 526 annotated images with over 1.16 billion annotated pixels. Figure 6 shows examples of the semantic segmentation dataset. The segmentation maps use different colors to represent annotated classes, such as pedestrians, bicycles, vehicles, vegetation, and pathways, illustrating the variety of urban and

Table 3 Comparison with selected object detection datasets

Feature	AuRa (ours)	COCO	KITTI	Cityscapes	BDD100K
Images	6,349	118,000	7,481	5,000	100,000
Annotations	42,284	886,000	80,256	25,000	700,000
Pedestrian classes	4+	1	1	1	1
Vehicle diversity	High	Medium	High	Medium	High
Emergency vehicles	✓	✗	✗	✗	Limited
Bicycle/motorcycle detail	With/Without rider	✗	✗	✗	✗
Infrastructure objects	Detailed	Limited	Basic	Moderate	Basic
Annotation accuracy	Manual (3px)	Crowdsourced	Manual	Manual	Mixed
Application focus	Urban/Micromobility	General	Driving	Driving	Driving

**Fig. 6** Examples of semantic segmentation. Each block consists of pairs with the original image on top and the corresponding semantic segmentation below

mixed-use environments captured in the dataset. Multiple image pairs are arranged both vertically and horizontally to showcase different scenarios and perspectives. Table 4 provides a detailed breakdown of the annotated objects used in this dataset for semantic segmentation. The dataset includes a wide range of classes and subclasses, categorized based

on urban environments and common elements found in outdoor scenes. The number of annotated objects is measured in pixels, reflecting the total coverage of each class in the dataset.

Table 4 List and number of annotated objects for semantic segmentation

Class	Subclass	Number of annotated objects [px]	Sum
Traffic Lanes and Paths	Car lane	132,267,587	
	Pedestrian path	123,148,914	
	Bicycle lane	68,184,007	
	Shared bicycle-/pedestrian path	25,960,893	
	Unsealed path	20,919,407	
	Railway track	6,741,392	
	Parking space	3,162,453	
	Sealed free space	3,091,656	383,476,309
Vegetations	Vertical vegetation	240,426,471	
	Horizontal vegetation	41,790,469	282,216,940
Traffic-related Infrastructure	Vertical barrier	27,580,049	
	Curb stone	10,798,487	
	Lane boundary strips	3,390,177	
	Traffic sign	1,847,618	
	Traffic light	372,719	
	Centre line	125,416	44,114,466
Traffic participants	Vehicle	66,890,772	
	Pedestrian	11,424,698	
	Cyclist with bicycle	7,630,877	
	Bicycle without cyclist	2,500,074	
	Motorcycle without motorcyclist	5,083	
	Motorcyclist with motorcycle	0	88,451,504
Miscellaneous	Sky	106,527,459	
	Miscellaneous static object	263,887,104	
	Miscellaneous dynamic objects	75,060	370,489,623
			1,168,748,842

The values highlighted in bold are the sum of all subclasses or the corresponding main class

- *Traffic Lanes and Paths* encompass various types of paths and lanes that are crucial for traffic flow and transportation in urban areas. The dataset annotates large portions of car lanes (132.3 million pixels) and pedestrian paths (123.1 million pixels), highlighting their prominence in the scene. Lesser-used paths like bicycle lanes and shared pedestrian-bicycle paths are also annotated, with around 68.2 million and 26 million pixels, respectively. Parking spaces, railway tracks, and unsealed paths are represented as well, ensuring a comprehensive annotation of infrastructure elements.
- *Vegetation* is a significant focus within the dataset, highlighting natural elements in the scene. Horizontal vegetation, such as grass or low shrubs, accounts for over 41 million annotated pixels, while vertical vegetation, which includes trees and taller plants, dominates with 240 million pixels. This distinction is vital for the accurate segmentation of urban green spaces and natural barriers.
- *Traffic-Related Infrastructure* includes key elements of the urban landscape, such as curbstones, lane boundary strips, and vertical barriers, which are meticulously annotated. Curbstones account for 10.7 million pixels, while lane boundary strips cover 3.4 million pixels.
- Vertical barriers, including structures like fences and guardrails, contribute over 27.6 million pixels, illustrating their importance in defining the road infrastructure. Additionally, traffic signs and lights, though smaller in pixel coverage, are essential for traffic regulation and safety, with 1.8 million and 0.3 million pixels, respectively.
- *Traffic Participants* are a wide array of annotations, including pedestrians (11.4 million pixels), vehicles (66.9 million pixels), and cyclists (7.6 million pixels). These annotations allow the dataset to distinguish between different types of road users, including cyclists with and without bicycles, as well as motorcyclists, though the latter group is sparsely represented with only 7,475 pixels.
- The *Miscellaneous* category captures objects that do not neatly fit into the other categories but are still common in urban scenes. The sky, a significant background element, is represented by over 102.8 million pixels. Other miscellaneous static objects, such as benches, trash bins, or construction materials, account for 25.9 million pixels, while dynamic objects, which could include temporary elements like moving obstacles, account for 126,380 pixels.

Table 5 Comparison with selected semantic segmentation datasets

Feature	AuRa (ours)	Cityscapes	TAS500	Mapillary
Images	526	5,000	500	25,000
Resolution	2046×1086	2048×1024	2048×1024	up to 6K
Pixels (annot.)	1.17B	~1.18B	1.02B	N/A
Camera Setup	Front cam	Stereo front	360°	Mixed
Annotations	Semantic	Sem., Inst.	Sem., Inst., Pan.	Sem., Inst.
Class Struct.	5 cat., 23 cls.	7 cat., 30 cls.	5 cat., few cls.	8 cat., 66 cls.
Paths	Detailed (6 types)	Road, sidewalk	Road, sidewalk	Fine-grained
Vegetation	Two classes	Single class	Single class	Detailed
Infrastructure	Curbs, barriers, strips, signs	Signs, poles	Road marking, signs	Rich infra.
Participants	Veh., ped., cyc., moto.	Veh., ped., riders	Veh., ped., cyc.	Detailed incl. animals
Misc.	Sky, static/dyn. obj.	Sky, fence, wall	Sky, misc.	Clutter, sky, bldgs
Focus	Urban + mixed-use	Car-focused urban	Traffic scenes	Diverse global scenes

Fig. 7 Qualitative comparison of object detection results. From left to right: Original image with ground truth annotations, YOLOv8-XS prediction, YOLOv8-L prediction

4.4 Comparison with existing semantic segmentation datasets

To contextualize the contribution and structure of our dataset, we compare it with three widely used semantic segmentation benchmarks: Cityscapes, TAS500, and Mapillary Vistas. Table 5 provides a detailed comparison across key characteristics relevant to semantic segmentation in urban and mixed-use environments. Our dataset comprises 526 high-resolution images captured from a front-facing

camera, with over 1.17 billion annotated pixels. While the number of images is lower than Cityscapes (5,000) and Mapillary Vistas (25,000), the annotation density and granularity are noteworthy, particularly in categories relevant to micromobility and non-motorized infrastructure. Compared to Cityscapes and TAS500, which primarily focus on car-centric driving environments, our dataset emphasizes mixed-use and pedestrian zones by distinguishing between car lanes, pedestrian paths, bicycle lanes, shared-use paths, and unsealed trails. These fine-grained distinctions are

absent or coarsely represented in the other datasets. In contrast to the single-class vegetation annotation in Cityscapes and TAS500, we introduce a two-class distinction between vertical and horizontal vegetation to support finer environmental modeling. Similarly, traffic-related infrastructure is captured in greater detail, with separate annotations for curbstones, lane boundary strips, vertical barriers, centerlines, and sealed free spaces. Although Mapillary Vistas features a broader set of classes and more diverse scenes, its crowdsourced nature leads to highly variable resolutions and camera perspectives, making it less consistent for applications in mobile robotics. TAS500 offers surround-view coverage and supports panoptic segmentation, yet it includes fewer fine-grained class distinctions and is more aligned with autonomous driving use cases. Overall, our dataset complements existing resources by focusing on semantic segmentation for micromobility, mixed-use zones, and sidewalk-level detail—scenarios underrepresented in current benchmarks.

5 Baseline model and training configuration

5.1 Object detection

To establish a baseline for object detection performance on our proposed dataset, we employed the YOLOv8 architecture (Varghese et al. 2024), leveraging two pre-trained backbone variants: YOLOv8-XS and YOLOv8-L. These models were selected for their efficiency and robustness in detecting objects across varied scales. Specifically, we initialized the YOLO detectors using the YOLOV8Backbone from KerasCV (Matthew et al. 2024), pretrained on the COCO dataset (Lin et al. 2015). We adapted the YOLOv8 models to our dataset by configuring the number of classes to match the annotated object categories. The model architecture included a Feature Pyramid Network (FPN) with a depth of 3 to capture multi-scale object features. Training

was conducted with the Adam optimizer (Kingma and Ba 2014), employing a learning rate of 0.001 and a global gradient clipping threshold of 10.0. The loss functions comprised binary cross-entropy for classification and Complete Intersection over Union (CIoU) loss (Zheng et al. 2019)

for bounding box regression. Due to the limited size of the dataset, we applied data augmentation techniques such as horizontal flipping, random cropping, color jittering, and affine transformations to artificially increase the training data by a factor of two. This step was essential to improve generalization and increase robustness against overfitting. Training was performed for 300 epochs with a batch size of 1 on a single NVIDIA RTX 2080 GPU. To further mitigate overfitting, we implemented early stopping (Prechelt 1998), reducing the learning rate by a factor of 0.5 after three epochs without validation improvement, with a lower bound of 1×10^{-6} . Regular model checkpoints were saved during training using the ModelCheckpoint callback. Table 6 provides a detailed breakdown of the annotated object classes used in the object detection benchmark, showing the number of labeled instances per class across the training, validation, and test splits.

To assess the performance of our object detection models, we employed the widely adopted mean Average Precision (mAP) metric as defined by the COCO evaluation metrics (Lin et al. 2015). This metric computes mAP over a range of IoU thresholds (0.50 to 0.95, in steps of 0.05) and object sizes (small, medium, large), as well as average recall under different detection limits. The results for the YOLOv8-XS and YOLOv8-L models are summarized in Table 7. Although the overall mAP scores are modest, particularly for smaller object sizes, we observe that the larger model variant (YOLOv8-L) consistently outperforms the smaller one across most evaluation metrics, especially for medium-sized objects.

Table 6 Object class distribution of training, validation and test data

Class	Training	Validation	Test
Car	13,339	1483	1503
Bus	87	10	5
Tram	597	67	69
Transporter	1824	203	256
Truck	605	68	65
Biker	3451	384	436
Emergency Vehicles	72	9	10
Traffic Lights/Railroad Crossing Signals	1626	181	185
Person	8375	931	1195
Dog	42	5	7
Traffic Signs	5027	559	577
Speed Bump/Traffic Markings	277	31	40
Sum	35,322	3931	4348

The values highlighted in bold are the sums of all classes for the training/validation/test data

Table 7 COCO-style evaluation metrics for YOLOv8-XS and YOLOv8-L

Metric	YOLOv8-XS [%]	YOLOv8-L [%]
mAP@[0.5:0.95]	14.4	17.2
mAP@0.5	20.2	23.2
mAP@0.75	16.8	20.1
mAP (small objects)	3.1	10.4
mAP (medium objects)	16.7	19.7
mAP (large objects)	18.0	23.3
Recall@1	11.7	13.4
Recall@10	16.8	20.1
Recall@100	16.8	20.2
Recall (small objects)	3.9	11.2
Recall (medium objects)	18.7	21.6
Recall (large objects)	21.1	27

5.2 Semantic segmentation

To establish a baseline for semantic segmentation performance on our proposed dataset, we employed a high-resolution network (HRNet) architecture, which is known for maintaining high-resolution representations throughout the model pipeline. Specifically, we utilized the pre-trained HRNetV2-W48 model, available via TensorFlow Hub (Abadi et al. 2015), originally trained on the ADE20K dataset (Zhou et al. 2019). We adapted the HRNet model to our dataset by appending a Conv2D layer with a 1×1 kernel and softmax activation to project the output to the desired number of segmentation classes. The full model was compiled using the Adam optimizer (Kingma and Ba 2014) and trained end-to-end with the Online Hard Example Mining (OHEM) loss (Shrivastava et al. 2016), which focuses learning on difficult samples to improve generalization in imbalanced data scenarios. Training was conducted for 150 epochs using a batch size of 1 on a single NVIDIA RTX 2080 GPU. To facilitate convergence and mitigate overfitting, we employed a learning rate scheduler based on validation loss. Specifically, the learning rate was reduced by a factor of 0.5 after three epochs without improvement, with a lower bound of 1×10^{-6} . The dataset was split into 422 training, 52 validation, and 52 test images, ensuring a balanced and consistent evaluation across all experiments. Evaluation was performed using two standard metrics for semantic segmentation: mean Pixel Accuracy (mPA) (Long et al. 2015) and mean Intersection over Union (mIoU) (Everingham et al. 2010). To further investigate the impact of class granularity on segmentation performance, we trained three variations of the baseline model using different class groupings derived from the full label set of 25 semantic classes. In the 19-class variant, several semantically similar classes were merged to reduce complexity. Specifically, pedestrian paths were combined with shared bicycle-/pedestrian paths, car lanes with parking spaces, and centre lines with lane boundaries.

Table 8 Segmentation performance of HRNet baseline models with different class granularities

Class granularity	mPA [%]	mIoU [%]
25 classes (all individual labels)	67.9	48.07
19 classes (partially merged)	69.36	57.13
13 classes (strongly merged)	76.7	67.2

Additionally, all bicycle- and motorcycle-related classes—both with and without riders (i.e., cyclist with bicycle, bicycle without cyclist, motorcyclist with motorcycle, and motorcycle without motorcyclist)—were grouped into a single class. The 13-class variant applies even broader semantic groupings through further merging. Bicycle lanes, pedestrian paths, shared bicycle-/pedestrian paths, sealed free spaces, and unsealed paths are combined into one class. Car lanes and parking spaces remain merged, as do vertical and horizontal vegetation. Centre lines and lane boundaries continue to be grouped together. The bicycle and motorcycle category remains unified, and traffic signs are merged with traffic lights. Finally, miscellaneous static and dynamic objects are consolidated into a single class (Table 8).

These results (Fig. 8) indicate that reducing class granularity improves segmentation performance, particularly in terms of mIoU. This is expected, as merging semantically or visually similar classes reduces label ambiguity and simplifies the classification task. However, such reductions may also obscure fine-grained distinctions critical for downstream applications, emphasizing a trade-off between performance and semantic detail.

6 Conclusions and future work

Despite the limited dataset size and class imbalance, our initial baseline experiments provide valuable insights. In the object detection task using YOLOv8, we observed modest mAP scores, which reflects the inherent challenges posed by a relatively small and imbalanced dataset. To partially mitigate this, we applied data augmentation techniques to artificially increase the training set size by a factor of two. Nevertheless, detection performance remains below that of benchmarks trained on large-scale datasets such as COCO. We interpret these results not as a limitation of the dataset's quality, but rather as an indicator of its current size and the complexity of its scenes. Importantly, this highlights the AuRa dataset's potential as a complementary resource for transfer learning or domain adaptation efforts, especially for urban navigation scenarios outside car-centric roadways. In contrast, semantic segmentation baselines achieved promising results. The HRNet-based models, trained on our densely annotated segmentation data, reached a mean IoU of up to 67.2% depending on class granularity. As expected,

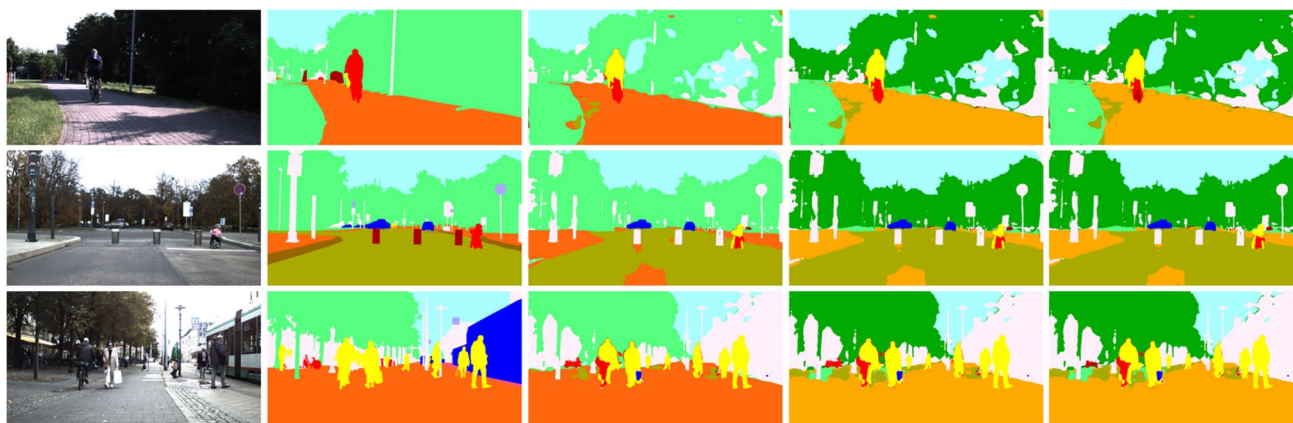


Fig. 8 Qualitative comparison of semantic segmentation results. Each column shows one example. The first row contains the original input images. The second row shows the ground truth annotations. Rows

three to five show the predictions from models trained with 13, 19, and all 25 classes, respectively

merging semantically similar labels improved performance by reducing label ambiguity. This trade-off between semantic detail and quantitative accuracy is essential to consider for downstream tasks such as navigation, scene understanding, and object-level reasoning. Taken together, these results underscore both the challenges and opportunities inherent in our dataset. While current baseline models yield room for improvement, especially in object detection, the dataset's richness and unique urban perspective make it a valuable foundation for research on micromobility-aware perception systems. We encourage the use of AuRa as a supplementary dataset in multi-dataset training setups, or as a target domain for domain adaptation strategies. As we expand the annotation set and enrich the sensor modalities (e.g., adding localization and motion data), we aim to further improve the dataset's utility for trajectory prediction, 3D perception, and multi-modal fusion research. The dataset is freely available for non-commercial research purposes only. Access can be requested at https://github.com/ovgu-mtk/aura_dataset/.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s41315-025-00502-x>.

Acknowledgements The project Aura was funded by the federal state Saxony-Anhalt and the European Regional Development Funds (EFRE, 2014–2020), project number 19-15003/004.

Author Contributions St.S and M.H. wrote the main manuscript and St.S, M.H. and J.W. supervised the annotation process St.S created the helper scripts All authors reviewed the manuscript.

Funding Open Access funding enabled and organized by Projekt DEAL.

Data Availability The dataset will be made available on our university website (<https://www.aura.ovgu.de>) in the future. it can be requested in advance by e-mail from the authors.

Declarations

Competing interests The authors declare no competing interests.

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